Roll No. | | | | | | | | | | | | Total No. of Pages : 02

Total No. of Questions : 13

B.Pharma (Sem.-4)

PHARMACEUTICAL ORGANIC CHEMISTRY-III

Subject Code : BP-401 T

M Code : 75843

Date of Examination : 18-12-2025

Time : 3 Hrs. Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.**
3. **SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.**

SECTION-A

1. **Write short notes on:**

   a) What is *Syn* and *anti* system of nomenclature?

   b) What is a meso compound? Give any example.

   c) Draw the structure of azete, imidazole, Isoxazole and quinoline.

   d) What is Clemmenson Reduction?

   e) Define Atropisomerism. Give examples.

   f) Differentiate between D and L configuration.

   g) Give any two reactions of Furan.

   h) Write down the structure and medicinal use of azepine.

   i) Define absolute asymmetric synthesis.

   j) Write down the steps involved in synthesis of indole.

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SECTION - B

2. Discuss in detail about the various methods of synthesis and chemical reactions of furan.

3. What is the requirement of optical activity? Give reasons. Comment in detail upon the various elements of symmetry.

4. **Comment upon synthesis of following heteronuclear compounds :**

   a) Pyrimidine

   b) Imidazole

   c) Pyrrole

   d) Isoquinoline.

SECTION - C

5. Write a detailed note on Dakin reaction and Wolf Kischner reduction reactions.

6. Give the comparative note on aromaticity of pyrrole, furan and thiophene.

7. What is conformational isomerism? Comment upon the isomerism present in n-butane.

8. Explain the concept and nomenclature of geometrical isomerism with the help of examples.

9. Write a detailed note on stereospecific reactions.

10. Comment upon the reactions and medicinal uses of Imidazole and Thiazole.

11. What are CIP rules? Explain in detail.

12. Write down the various reactions of Pyridine.

13. Write a detailed note on the reactions of chiral compounds.

NOTE : Disclosure of Identity by writing Mobile No. or Making of passing request on any page of Answer Sheet will lead to UMC against the Student.